

Meat intake and mortality: a prospective study of over half a million people



Context High intakes of red or processed meat may increase risk of mortality. Objective Determine the relations of red, white and processed meat intakes to risk for total, and cause-specific mortality. Design, Setting, and Participants The NIH-AARP Diet and Health Study cohort of half a million people aged 50-71 years at baseline. Meat intake was estimated from a food frequency questionnaire administered at baseline. Cox proportional hazards regression estimated hazard ratios (HRs) and 95% confidence intervals (CIs) within quintiles of meat intake. The covariates included in the models were: age; education; marital status; family history of cancer (yes/no) (cancer mortality only); race; body mass index; 31-level smoking history; physical activity; energy intake; alcohol intake; vitamin supplement use; fruit consumption; vegetable consumption; and menopausal hormone therapy among women. Main Outcome Measure Total mortality, deaths due to cancer, CVD, accidents, and other causes. Results There were 47,976 male deaths and 23,276 female deaths during 10 years of follow-up. Men and women in the highest versus lowest quintile of red (HR 1.31, 95% CI 1.27-1.35; HR 1.36, 95% CI 1.30-1.43, respectively) and processed meat intake (HR 1.16, 95% CI 1.12-1.20; HR 1.25, 95% CI 1.20-1.31, respectively) had elevated risks for overall mortality. Regarding cause-specific mortality, men and women had elevated risks for cancer mortality for red (HR 1.22, 95% CI 1.16-1.29; HR 1.20, 95% CI 1.12-1.30, respectively) and processed meats (HR 1.12, 95% CI 1.06-1.19; HR 1.11, 95% CI 1.04-1.19, respectively). Furthermore, CVD risk was elevated for men and women in the highest quintile of red (HR 1.27, 95% CI 1.20-1.35; HR 1.50, 95% CI 1.37-1.65, respectively) and processed meat (HR 1.09, 95% CI 1.03-1.15; HR 1.38, 95% CI 1.26-1.51, respectively). When comparing the highest to the lowest quintile of white meat intake, there was an inverse association for total mortality, and cancer mortality, as well as all other deaths for both men and women. Conclusion Red and processed meat intakes were associated with modest increases in total mortality, cancer mortality and CVD mortality.